

Verification o6_H_Simulator_FVM_1D_CompressibleRock

Barnaby Fryer

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical steady-state solution and transient solution for one-dimensional slightly compressible single-phase flow in a porous medium with non-constant porosity and permeability.

2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a heterogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x} \left(\frac{\rho k}{\mu} \frac{\partial P}{\partial x} \right) + \frac{Q}{V}, \quad (1)$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m³],
- ϕ is porosity [-],
- k is permeability [m²],
- Q represents source terms [kg/s],
- V is the cell volume [m³],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho} \frac{\partial \rho}{\partial P}, \quad (2)$$

the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \quad (3)$$

where ρ_0 is the reference density at pressure P_0 . Similarly, ϕ and k are found by,

$$\phi = \phi_0 e^{c_\phi(P-P_0)}, \quad k = k_0 e^{c_k(P-P_0)}. \quad (4)$$

3 Analytical Solution

3.1 Permeability

To validate the numerical simulator with time-varying permeability, a steady-state solution is used. In steady state, the flux is constant and accumulation constant. As the flux is constant everywhere, it follows that the Darcy flux is given locally for two adjacent cells by,

$$Q = -\frac{k\rho}{\mu} \frac{dP}{dx} A = -\frac{k\rho}{\mu} \frac{P_2 - P_1}{x_2 - x_1} A. \quad (5)$$

For these two adjacent cells, the pressure at the second cell, P_2 can therefore be calculated based on the pressure in the first cell, P_1 , as

$$P_2 = P_1 - \frac{Q\mu(x_2 - x_1)}{Ak\rho}. \quad (6)$$

After this pressure update, k must be updated. If the fluid compressibility, c_f is sufficiently small, the effect of a changing $\rho \approx \rho_0$ is negligible. With the new value of k , the pressure calculation is repeated. This process continues until pressure converges.

3.2 Porosity

To validate that mass is still being conserved despite a changing porosity, a transient simulation is used. Mass is injected at a constant rate and porosity is allowed to evolve. The system mass is summed and compared to the injected mass to ensure mass was neither created nor destroyed.

4 Simulation Parameters

The simulation parameters are given in Table 1 for the steady-state permeability verification and Table 2 for the transient porosity verification.

Parameter	Symbol	Value
Number of cells	N_x	500
Domain length	L_x	10 m
Initial permeability	k_0	$1 \times 10^{-11} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Initial porosity	ϕ_0	0.2
Fluid compressibility	c_f	$1 \times 10^{-10} \text{ Pa}^{-1}$
Permeability “compressibility”	c_k	$3 \times 10^{-8} \text{ Pa}^{-1}$
Rock compressibility	c_ϕ	$1 \times 10^{-9} \text{ Pa}^{-1}$
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$
Well pressures	P_w	$10 \times 10^6 \text{ Pa}$
Well locations	x_w	0 m
Wellbore indices	WI	$1 \times 10^7 \text{ m}$
Well mass rates	Q_w	1 kg/s
Constant mass-rate well locations	x_m	10 m

Table 1: Simulation parameters for permeability test

Parameter	Symbol	Value
Number of cells	N_x	500
Domain length	L_x	10 m
Initial permeability	k_0	$1 \times 10^{-11} \text{ m}^2$
Viscosity	μ	0.1 Pa·s
Initial porosity	ϕ_0	0.2
Fluid compressibility	c_f	$1 \times 10^{-10} \text{ Pa}^{-1}$
Permeability “compressibility”	c_k	$1 \times 10^{-8} \text{ Pa}^{-1}$
Rock compressibility	c_ϕ	$1 \times 10^{-8} \text{ Pa}^{-1}$
Reference density	ρ_0	1000 kg/m ³
Reference pressure	P_0	$1 \times 10^5 \text{ Pa}$
Well mass rates	Q_w	1 kg/s
Constant mass-rate well locations	x_m	10 m

Table 2: Simulation parameters for constant mass rate well test.

5 Results

5.1 Permeability check

Using the parameters in Table 1, the pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

5.2 Porosity check

For the parameters, given in Table 2, both the MATLAB and Python simulator show mass conservation, Figure 5 and Figure 6, despite significant changes in porosity, Figure 3 and Figure 4. Note the values of porosity differ between the two simulators because the pressure differs. The pressure differs because there is no boundary condition fixing the pressure in this case (as there would be either with a Dirichlet boundary condition at an edge or a fixed pressure well). What is important is the large change in porosity and the mass being conserved.

6 Discussion

The numerical pressure distribution reproduces the analytical pressure solution at steady state and conserves mass during transient pressure changes.

7 Conclusion

The simulator successfully reproduces the analytical steady-state solution and transient solution for one-dimensional slightly compressible flow with varying porosity and permeability. This provides verification that the finite volume implementation is correctly solving the governing equations for these benchmark problems.

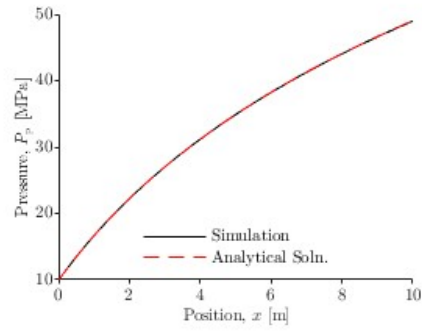


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator for constant pressure wells.

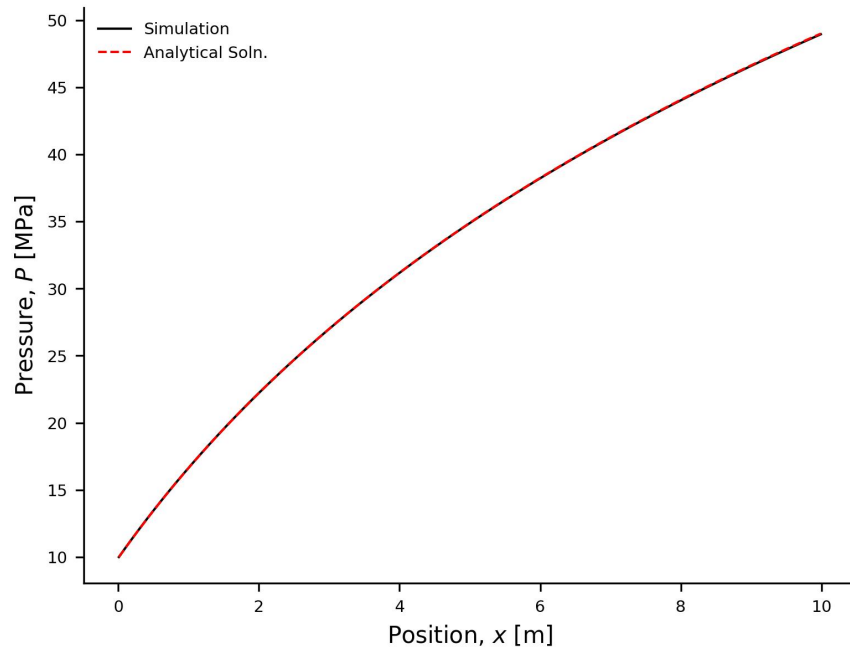


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator for constant pressure wells.

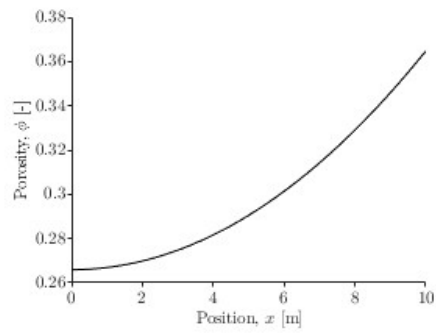


Figure 3: The final state of porosity for the simulation for the MATLAB simulator.

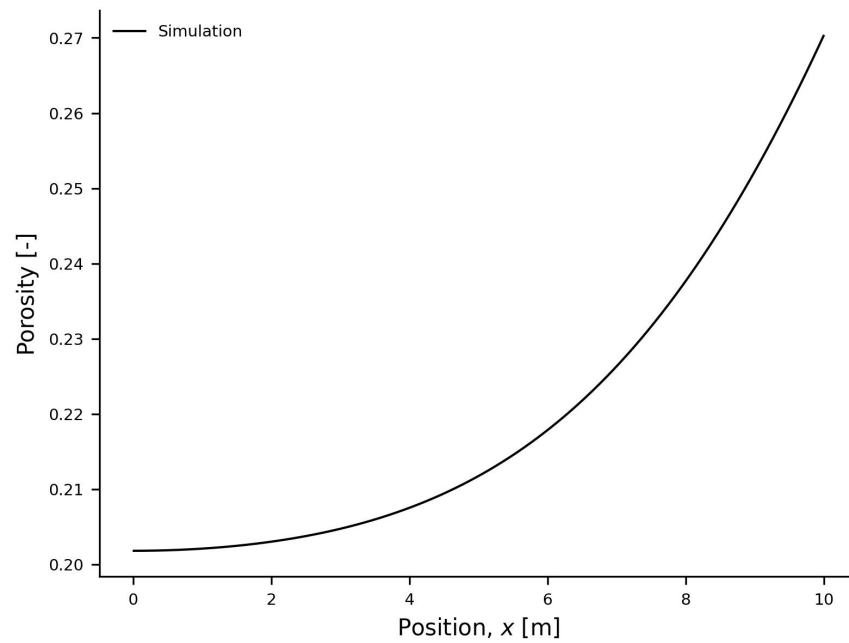


Figure 4: The final state of porosity for the simulation for the Python simulator.

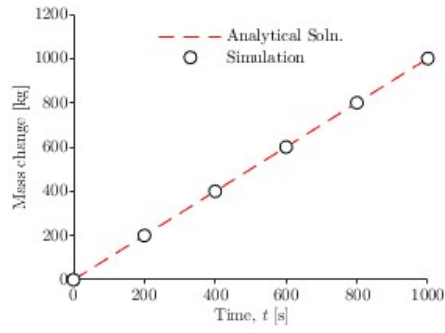


Figure 5: The total mass of the system from the simulation versus the injected mass for the MATLAB simulator. Note that mass has been conserved despite an evolving porosity.

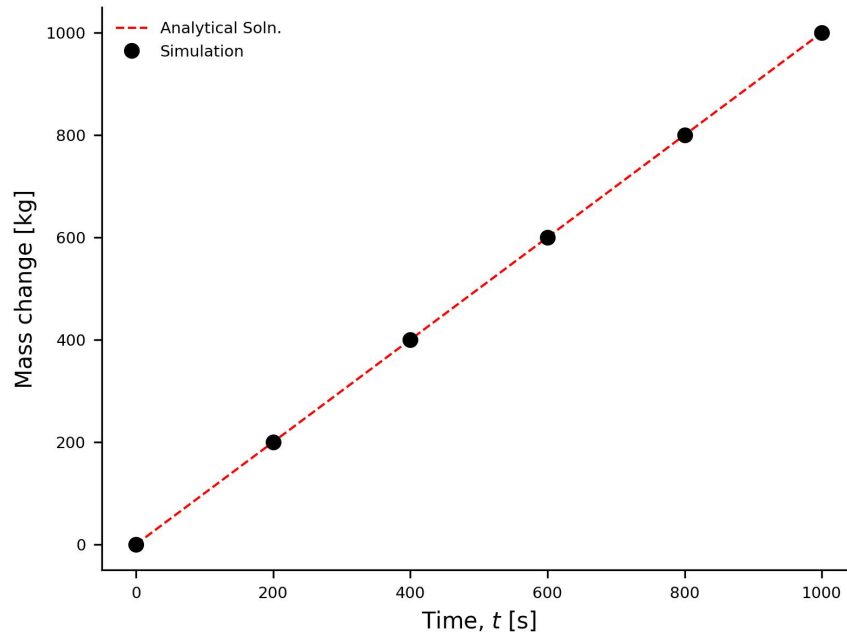


Figure 6: The total mass of the system from the simulation versus the injected mass for the Python simulator. Note that mass has been conserved despite an evolving porosity.